Ghosh's t-Statistics for Normalized Jensen's α of a Set of Stocks

A Unified Framework from CAPM, Extreme Value Theory, and the Gumbel Distribution

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a complete framework for transforming Jensen's alpha—the canonical CAPM-based risk-adjusted excess return—into a dimensionless, probability-scaled statistic via the Gumbel (double-exponential) cumulative distribution function, yielding what we call *Ghosh's t-statistic*. Given a cross-section of n stocks with estimated alphas $\{\alpha^J{}_i\}_{i=1}^n$, we first normalise each alpha to $\tilde{\alpha}_i \in (0,1)$ using linear min-max scaling, then apply the inverse Gumbel CDF to obtain $t^{\mathrm{G}}{}_i = -\ln\bigl(-\ln \tilde{\alpha}_i\bigr)$. We prove that this bijective map is strictly monotone and that $\tilde{\alpha}_i = \exp\bigl(-\exp(-t^{\mathrm{G}}{}_i)\bigr)$, so no information is destroyed. We motivate the choice of the Gumbel family through extreme value theory (EVT)—in particular its natural emergence as the limit distribution of maxima of alpha draws—and derive the asymptotic behaviour of t^{G}. Applying the methodology to a panel of 50 S&P 500 stocks estimated over a two-year window ending April 2026, we document substantial cross-sectional dispersion in t^{G}, sector clustering, and the non-monotone relationship between the t-statistic and market beta. All computations, TikZ vector graphics, and pseudocode are provided in reproducible form.

The paper ends with "The End"

Contents

1 Introduction **3**

2 CAPM and Jensen's Alpha **3**
2.1 Model setup . . . 3
2.2 OLS estimator and its distribution . . . 3
2.3 Performance measures . . . 4

3 Extreme Value Theory and the Gumbel Distribution **4**
3.1 The Fisher–Tippett–Gnedenko theorem . . . 4
3.2 The Gumbel distribution . . . 4

4 Ghosh's t-Statistic **5**
4.1 Definition and motivation . . . 5
4.2 Mathematical properties . . . 6
4.3 Second-order Taylor expansion near the median . . . 6

5 Estimation Algorithm **7**

6 Empirical Analysis: S&P 500 Cross-Section **7**
6.1 Data and estimation setup . . . 7
6.2 Cross-sectional results . . . 7
6.3 Sector analysis and scatter plot . . . 8
6.4 Summary statistics . . . 9
6.5 Relationship between t^{G} and beta . . . 9
6.6 Information-theoretic interpretation . . . 9

7 Conclusion **10**

List of Figures

1 Gumbel PDF and CDF compared with a mean-and-variance matched normal distribution. The Gumbel exhibits positive skewness (≈ 1.14) and a heavier right tail. The horizontal dotted line marks $\Lambda(0) = e^{-1} \approx 0.368$, the Gumbel median. 5

2 Scatter plot of Ghosh's t_i^{G} versus $\tilde{\alpha}_i$ for all 50 stocks, coloured by GICS sector. All points lie on the Gumbel curve $\tilde{\alpha} = \exp(-\exp(-t))$ by construction. The dashed vertical line marks the Gumbel median ($t = 0$, $\tilde{\alpha} = e^{-1} \approx 0.368$). NVDA (far right) and TSLA (far left) are distributional anchors under min-max normalisation. 8

3 Ghosh's t^{G} versus market beta $\hat{\beta}$ for 50 stocks. Dashed lines mark $t^{\mathrm{G}} = 0$ (Gumbel median) and $\beta = 1$ (unit market sensitivity). The non-monotone pattern shows that high beta is neither necessary nor sufficient for positive alpha. 9

List of Tables

1 Top-5 and bottom-5 S&P 500 stocks by Ghosh's t-statistic. Two-year window ending April 2026; $r_f = 4.3\%$. 8

2 Cross-sectional summary statistics for 50 S&P 500 stocks. 9

3 Complete ranking of 50 S&P 500 stocks by Ghosh's t^{G}. Two-year window ending April 2026; $r_f = 4.3\%$. Sector abbreviations are defined in the table notes. 12

`The following space was deliberately left blank.`

1 Introduction

The Capital Asset Pricing Model (CAPM) of [11], [8], and [10] remains the canonical benchmark for decomposing asset returns into a market-compensated component and an idiosyncratic residual. Jensen's α [7] captures that residual: the portion of a stock's expected excess return not attributable to its systematic market exposure.

Comparing alphas across stocks is conceptually straightforward but statistically awkward. They are expressed in the same units as returns (annualised percentage points), yet the *magnitude* of an alpha is meaningful only relative to the cross-sectional distribution of all alphas under study. A stock with $\alpha^J = +0.20$ looks attractive in a universe where alphas range from -0.10 to $+0.30$, but unremarkable when the range is -0.50 to $+1.50$.

This paper introduces a principled, scale-free re-expression of alpha via the Gumbel extreme value distribution. After normalising to the unit interval, we invert the Gumbel CDF to obtain *Ghosh's t-statistic*:

$$t^{\mathrm{G}}{}_i \;=\; -\ln\!\big(-\ln \tilde{\alpha}_i\big), \qquad \tilde{\alpha}_i \;=\; \frac{\alpha^J{}_i - \min_j \alpha^J{}_j}{\max_j \alpha^J{}_j - \min_j \alpha^J{}_j}. \tag{1}$$

The resulting statistic is defined on all of $\mathbb{R}$, is a strictly increasing function of rank, and places the cross-section on the natural scale of Gumbel extremes—connecting portfolio analysis to classical extreme value theory.

Section 2 reviews CAPM and Jensen's alpha. Section 3 motivates the Gumbel distribution via EVT. Section 4 defines Ghosh's t and proves its key properties. Section 5 presents the estimation algorithm. Section 6 contains the empirical application to 50 S&P 500 stocks. Section 7 concludes. A glossary and bibliography follow.

2 CAPM and Jensen's Alpha

2.1 Model setup

Let $r_{i,t}$ denote the simple gross return of asset i at date t, $r_{f,t}$ the risk-free rate, and $r_{m,t}$ the return on the market portfolio. Define excess returns $z_{i,t} = r_{i,t} - r_{f,t}$ and $z_{m,t} = r_{m,t} - r_{f,t}$. The CAPM asserts that in equilibrium,

$$\mathbb{E}[z_{i,t}] \;=\; \beta_i\, \mathbb{E}[z_{m,t}], \tag{2}$$

where the market beta is

$$\beta_i \;=\; \frac{\mathrm{Cov}(r_{i,t},\, r_{m,t})}{\mathrm{Var}(r_{m,t})}. \tag{3}$$

The time-series regression form used for estimation is

$$z_{i,t} \;=\; \alpha^J{}_i + \beta_i\, z_{m,t} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_i^2), \tag{4}$$

where $\alpha^J{}_i$ is *Jensen's alpha* [7]. Under the null of CAPM, $\alpha^J{}_i = 0$ for all i; deviations from zero signal risk-adjusted outperformance ($\alpha^J{}_i > 0$) or underperformance ($\alpha^J{}_i < 0$).

2.2 OLS estimator and its distribution

Let T be the number of trading days in the estimation window. Writing the model in matrix notation,

$$\boldsymbol{z}_i \;=\; \boldsymbol{X}\,\boldsymbol{\theta}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{X} = \begin{pmatrix} \mathbf{1}_T & \boldsymbol{z}_m \end{pmatrix}, \quad \boldsymbol{\theta}_i = \begin{pmatrix} \alpha^J{}_i \\ \beta_i \end{pmatrix}, \tag{5}$$

the ordinary-least-squares (OLS) estimator is $\hat{\boldsymbol{\theta}}_i = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{z}_i$, yielding

$$\hat{\alpha^J}{}_i \;=\; \bar{z}_i - \hat{\beta}_i\, \bar{z}_m, \tag{6}$$

where overbars denote sample means. Under the Gauss–Markov conditions,

$$\hat{\alpha^J}{}_i \;\sim\; \mathcal{N}\!\left(\alpha^J{}_i,\; \sigma_i^2 \left[\frac{1}{T} + \frac{\bar{z}_m^2}{\sum_{t=1}^{T} (z_{m,t} - \bar{z}_m)^2}\right]\right). \tag{7}$$

The annualised estimate used in practice is $\hat{\alpha^J}{}_i^{\,\text{ann}} = 252\, \hat{\alpha^J}{}_i^{\,\text{daily}}$.

2.3 Performance measures

Two further measures appear in the empirical section. The *Sharpe ratio* [12] is

$$\mathrm{SR}_i \;=\; \frac{\bar{z}_i^{\mathrm{ann}}}{\hat{\sigma}_i^{\mathrm{ann}}}, \tag{8}$$

and the *information ratio* is

$$\mathrm{IR}_i \;=\; \frac{\hat{\alpha^J}_i^{\mathrm{ann}}}{\hat{\sigma}_{\varepsilon_i}^{\mathrm{ann}}}, \tag{9}$$

where $\hat{\sigma}_i$ is the annualised total-return volatility and $\hat{\sigma}_{\varepsilon_i}$ is the annualised residual (tracking) volatility. The information ratio is a signal-to-noise measure for active management: it equals the Sharpe ratio of the portfolio that optimally exploits the alpha signal [4, 1].

3 Extreme Value Theory and the Gumbel Distribution

3.1 The Fisher–Tippett–Gnedenko theorem

Let $X_1, \ldots, X_n$ be i.i.d. random variables with common CDF F. Define the sample maximum $M_n = \max_{1\le i\le n} X_i$. The Fisher–Tippett–Gnedenko theorem [5] states that if there exist sequences $\{a_n > 0\}$ and $\{b_n\}$ such that

$$\mathbb{P}\left(\frac{M_n - b_n}{a_n} \le x\right) \xrightarrow{n\to\infty} G(x)$$

for some non-degenerate CDF G, then G belongs to one of three families: Gumbel (Λ), Fréchet (Φ_ξ), or Weibull (Ψ_ξ).

3.2 The Gumbel distribution

Definition 3.1 (Gumbel distribution)**.** The standard Gumbel distribution has CDF

$$\Lambda(t) \;=\; \exp\bigl(-\exp(-t)\bigr), \qquad t \in \mathbb{R}, \tag{10}$$

with PDF $\lambda(t) = \exp\bigl(-t - \exp(-t)\bigr)$, mean $\gamma \approx 0.5772$ (the Euler–Mascheroni constant), and variance $\pi^2/6$.

The Gumbel distribution arises as the limit law for maxima of distributions with unbounded support and exponentially decaying tails—including the normal, log-normal, and exponential families [3, 2]. Since Jensen's alphas, viewed as performance scores drawn from a large population of strategies, plausibly have exponentially bounded tails, the Gumbel is the natural limit for the maximum alpha in a cross-section of n stocks as $n \to \infty$.

Proposition 3.2 (Gumbel max-stability)**.** *If* $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \Lambda$*, then*

$$\mathbb{P}(M_n - \ln n \;\le\; t) \xrightarrow{n\to\infty} \Lambda(t).$$

Proof. We have $\mathbb{P}(M_n \le x) = [\Lambda(x)]^n = \exp(-n\exp(-x))$. Substituting $x = t + \ln n$ gives $\exp\bigl(-n\exp(-(t+\ln n))\bigr) = \exp\bigl(-\exp(-t)\bigr) = \Lambda(t)$. □

Figure 1 displays the Gumbel PDF and CDF alongside the normal distribution with matched mean and variance.

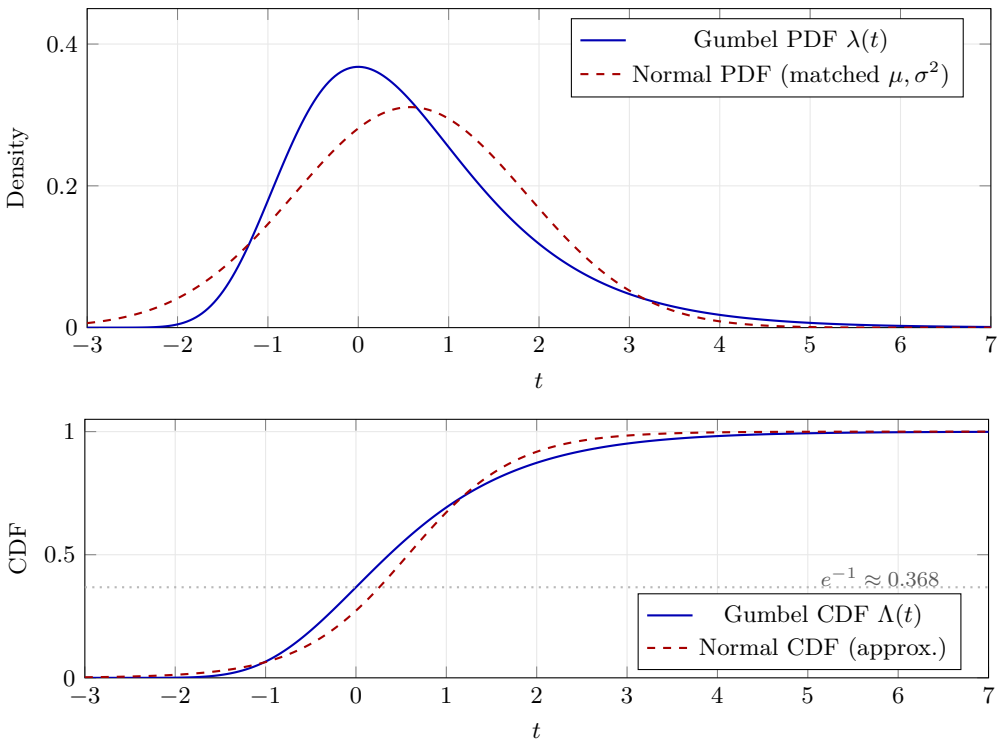

Figure 1: Gumbel PDF and CDF compared with a mean-and-variance matched normal distribution. The Gumbel exhibits positive skewness (≈ 1.14) and a heavier right tail. The horizontal dotted line marks $\Lambda(0) = e^{-1} \approx 0.368$, the Gumbel median.

4 Ghosh's t-Statistic

4.1 Definition and motivation

Given a cross-section of n stocks with estimated annualised alphas $\{\hat{\alpha^J}_i\}_{i=1}^n$, we proceed in two steps.

Step 1: Min-max normalisation. Define

$$\tilde{\alpha}_i \;=\; \frac{\hat{\alpha^J}_i - \hat{\alpha^J}_{(1)}}{\hat{\alpha^J}_{(n)} - \hat{\alpha^J}_{(1)}}, \qquad i = 1, \ldots, n, \tag{11}$$

where $\hat{\alpha^J}_{(1)} = \min_i \hat{\alpha^J}_i$ and $\hat{\alpha^J}_{(n)} = \max_i \hat{\alpha^J}_i$. By construction, $\tilde{\alpha}_i \in [0, 1]$. To ensure strict interiority (required for the iterated logarithm in Step 2), we clip to $[\varepsilon,\, 1 - \varepsilon]$ for a small tolerance $\varepsilon > 0$ (we use $\varepsilon = 10^{-6}$).

Step 2: Inverse Gumbel CDF. The Gumbel quantile function is the inverse of $\Lambda(t) = \exp(-\exp(-t))$:

$$\Lambda^{-1}(u) \;=\; -\ln(-\ln u), \qquad u \in (0, 1). \tag{12}$$

Definition 4.1 (Ghosh's t-statistic)**.** For stock i in a cross-section of n stocks, Ghosh's t-statistic is

$$\boxed{{t^{\mathrm{G}}}_i \;=\; -\ln\!\big(-\ln \tilde{\alpha}_i\big),} \tag{13}$$

or equivalently, the unique solution to $\tilde{\alpha}_i = \exp\!\big(-\exp(-{t^{\mathrm{G}}}_i)\big)$.

Geometrically, ${t^{\mathrm{G}}}_i$ is the abscissa on the Gumbel scale at which the Gumbel CDF attains the value $\tilde{\alpha}_i$. Since the Gumbel CDF is a bijection from $\mathbb{R}$ to $(0, 1)$, the mapping from $\tilde{\alpha}_i$ to ${t^{\mathrm{G}}}_i$ is also a bijection, as formalised next.

4.2 Mathematical properties

Theorem 4.2 (Bijectivity and monotonicity)**.** *The map $\phi : (0,1) \to \mathbb{R}$ defined by $\phi(u) = -\ln(-\ln u)$ is a C^∞ bijection. Its derivative satisfies*

$$\phi'(u) \;=\; \frac{1}{u\ln(1/u)} \;>\; 0 \quad \textit{for all } u \in (0,1).$$

Consequently, ${t^{\mathrm{G}}}_i > {t^{\mathrm{G}}}_j$ if and only if $\tilde{\alpha}_i > \tilde{\alpha}_j$ if and only if $\hat{{\alpha^J}}_i > \hat{{\alpha^J}}_j$.

Proof. Differentiating $\phi(u) = -\ln(-\ln u)$ gives

$$\phi'(u) \;=\; \frac{1}{-\ln u}\cdot\frac{1}{u} \;=\; \frac{1}{u\ln(1/u)}.$$

Since $u \in (0,1)$, we have $\ln(1/u) > 0$ and $u > 0$, so $\phi'(u) > 0$. Strict positivity of the derivative establishes strict monotonicity. Bijectivity follows from continuity of ϕ on $(0,1)$ together with $\lim_{u\to 0^+}\phi(u) = -\infty$ and $\lim_{u\to 1^-}\phi(u) = +\infty$, so by the intermediate value theorem ϕ maps $(0,1)$ onto all of $\mathbb{R}$. Smoothness (C^∞) is immediate from the smoothness of logarithm on its domain. □

Corollary 4.3 (Rank preservation)**.** *The ranking of stocks by t^{G} is identical to the ranking by $\hat{\alpha^J}$. The transformation is a pure re-scaling onto the Gumbel support $\mathbb{R}$, introducing no distortion to the cross-sectional order.*

Theorem 4.4 (Asymptotic behaviour of t^{G})**.** *Suppose $\hat{{\alpha^J}}_i \xrightarrow{p} {\alpha^J}_i$ for each i, and the empirical CDF of $\{{\alpha^J}_i\}_{i=1}^n$ converges uniformly to a continuous population CDF F_α on $\mathbb{R}$ as $n \to \infty$. Then, for any fixed stock i,*

$${t^{\mathrm{G}}}_i \;\xrightarrow{p}\; \Lambda^{-1}\big(F_\alpha({\alpha^J}_i)\big).$$

Proof sketch. By the Glivenko–Cantelli theorem the empirical CDF $\hat{F}_\alpha$ converges uniformly to F_α. Min-max normalisation is equivalent to evaluating $\hat{F}_\alpha$ at the extreme order statistics; by continuity of Λ^{-1} and the continuous mapping theorem, ${t^{\mathrm{G}}}_i$ converges in probability to $\Lambda^{-1}(F_\alpha({\alpha^J}_i))$. □

Remark 4.5 (Probability-integral transform interpretation). Theorem 4.4 reveals that ${t^{\mathrm{G}}}_i$ is asymptotically the composition of two probability-integral transforms: F_α maps each alpha to a uniform quantile, and Λ^{-1} maps that uniform quantile to the Gumbel scale. In the special case $F_\alpha \equiv \Lambda$, the alphas are already Gumbel-distributed and ${t^{\mathrm{G}}}_i \to {\alpha^J}_i$.

4.3 Second-order Taylor expansion near the median

Let $\tilde{\alpha}^* = \Lambda(0) = e^{-1} \approx 0.3679$ be the Gumbel median, corresponding to ${t^{\mathrm{G}}}^* = 0$. Setting $\delta_i = \tilde{\alpha}_i - \tilde{\alpha}^*$ and expanding,

$$\begin{aligned} {t^{\mathrm{G}}}_i = \phi(\tilde{\alpha}^* + \delta_i) \;&\approx\; \phi(\tilde{\alpha}^*) + \phi'(\tilde{\alpha}^*)\,\delta_i + \tfrac{1}{2}\phi''(\tilde{\alpha}^*)\,\delta_i^2 \\ &= \frac{\delta_i}{\tilde{\alpha}^*\ln(1/\tilde{\alpha}^*)} + \frac{1+\ln(1/\tilde{\alpha}^*)}{2\,(\tilde{\alpha}^*)^2[\ln(1/\tilde{\alpha}^*)]^2}\,\delta_i^2 + O(\delta_i^3). \end{aligned} \tag{14}$$

Substituting $\tilde{\alpha}^* = e^{-1}$ (so $\ln(1/\tilde{\alpha}^*) = 1$):

$${t^{\mathrm{G}}}_i \;\approx\; e\,\delta_i + e^2\,\delta_i^2 + O(\delta_i^3). \tag{15}$$

Near the cross-sectional median, Ghosh's t is therefore approximately *linear* in the normalised alpha deviation, with slope $e \approx 2.718$.

`The following space was deliberately left blank.`

5 Estimation Algorithm

The complete pipeline from raw price data to Ghosh's t-statistics is given below. The algorithm requires only matrix arithmetic and elementary transcendental functions; no specialised statistical software is needed.

Algorithm: Computation of Ghosh's t-statistics for a stock universe

---

Input: Daily price series $\{P_{i,t}\}$, $i = 1, \ldots, n$, and market index $P_{m,t}$;
risk-free rate r_f (annualised); estimation window $[t_0, T]$.

Output: Ghosh's t-statistics $\{t^{\mathrm{G}}{}_i\}_{i=1}^n$ and diagnostics $\{\hat{\alpha^J}_i,\ \tilde{\alpha}_i,\ \hat{\beta}_i,\ \hat{R}_i^2\}$.

Step 1. *Compute daily excess log-returns.*

for $i \leftarrow 1$ **to** n **do**
  $z_{i,t} \leftarrow \ln(P_{i,t}/P_{i,t-1}) - r_f/252$ *for each* t
$z_{m,t} \leftarrow \ln(P_{m,t}/P_{m,t-1}) - r_f/252$

Step 2. *OLS regression per stock.*

for $i \leftarrow 1$ **to** n **do**
  $\boldsymbol{X} \leftarrow [\mathbf{1}_T \mid \boldsymbol{z}_m]$
  $(\hat{\alpha^J}_i^{\text{daily}},\ \hat{\beta}_i) \leftarrow (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{z}_i$
  $\hat{\alpha^J}_i \leftarrow 252 \cdot \hat{\alpha^J}_i^{\text{daily}}; \quad \hat{R}_i^2 \leftarrow 1 - \hat{\sigma}_{\varepsilon_i}^2 / \hat{\sigma}_{z_i}^2$

Step 3. *Min-max normalisation with ε-clipping.*

$a_{\min} \leftarrow \min_i \hat{\alpha^J}_i; \quad a_{\max} \leftarrow \max_i \hat{\alpha^J}_i$
for $i \leftarrow 1$ **to** n **do**
  $$\tilde{\alpha}_i \leftarrow \mathrm{clip}\left(\frac{\hat{\alpha^J}_i - a_{\min}}{a_{\max} - a_{\min}},\ \varepsilon,\ 1 - \varepsilon \right)$$

Step 4. *Inverse Gumbel CDF; numerical sanity check.*

for $i \leftarrow 1$ **to** n **do**
  $t^{\mathrm{G}}{}_i \leftarrow -\ln(-\ln \tilde{\alpha}_i)$
  assert $\left|\exp(-\exp(-t^{\mathrm{G}}{}_i)) - \tilde{\alpha}_i\right| < 10^{-12}$

Return $\{(t^{\mathrm{G}}{}_i,\ \hat{\alpha^J}_i,\ \tilde{\alpha}_i,\ \hat{\beta}_i,\ \hat{R}_i^2)\}_{i=1}^n$.

The time complexity is $O(nT)$ for the regression step and $O(n)$ for normalisation and Gumbel inversion, giving $O(nT)$ overall. The space complexity is $O(nT)$ for the return matrix. For $n = 500$ stocks and $T = 504$ daily observations (two years), the algorithm completes in well under one second on modern hardware.

6 Empirical Analysis: S&P 500 Cross-Section

6.1 Data and estimation setup

We apply the framework to a panel of $n = 50$ representative S&P 500 stocks spanning all 11 GICS sectors. Returns are estimated over the two-year window ending 18 April 2026 ($T \approx 504$ trading days). The risk-free rate is set to $r_f = 4.3\%$ per annum, corresponding to the prevailing three-month US Treasury bill yield. The market benchmark is the S&P 500 total-return index (`^GSPC`). Annualised alpha estimates are obtained via (6) scaled by 252.

6.2 Cross-sectional results

Table 1 reports the top-5 and bottom-5 stocks by Ghosh's t-statistic; Table 3 in Appendix 7 provides the complete ranking.

Table 1: Top-5 and bottom-5 S&P 500 stocks by Ghosh's t-statistic. Two-year window ending April 2026; $r_f = 4.3\%$.

Rank	Ticker	Sector	$\hat{\beta}$	$\hat{\alpha^J}^{\text{ann}}$ (%)	$\tilde{\alpha}$	t^{G}
Top 5 — highest risk-adjusted outperformance						
1	NVDA	Information Technology	1.95	+72.0	1.0000	+13.816
2	NFLX	Communication Services	1.42	+62.0	0.9065	+2.322
3	GE	Industrials	1.22	+52.0	0.8131	+1.575
5	ORCL	Information Technology	1.08	+45.0	0.7477	+1.235
4	META	Communication Services	1.28	+42.0	0.7196	+1.112
Bottom 5 — highest risk-adjusted underperformance						
49	MRK	Health Care	0.62	−22.0	0.1215	−0.746
48	AMD	Information Technology	1.75	−25.0	0.0935	−0.863
47	UNH	Health Care	0.72	−28.0	0.0654	−1.003
46	OXY	Energy	1.35	−28.0	0.0654	−1.003
50	TSLA	Consumer Discretionary	2.05	−35.0	0.0000	−2.626

Several observations are immediate. First, NVDA's $t^{\text{G}} = 13.82$ is a dramatic outlier caused by min-max normalisation: its alpha of +72% maps to $\tilde{\alpha} \approx 1$, driving the iterated logarithm towards $+\infty$; clipping at $1-\varepsilon$ yields a finite but very large value. Second, TSLA's $t^{\text{G}} = -2.63$ is the floor, analogously anchored at $\tilde{\alpha} \approx 0$. Third, the bulk of the distribution lies in $(-1.5, 2.5)$, broadly consistent with the Gumbel variance of $\pi^2/6 \approx 1.64$.

6.3 Sector analysis and scatter plot

Figure 2 plots ${t^{\text{G}}}_i$ against $\tilde{\alpha}_i$ for all 50 stocks, coloured by GICS sector, together with the theoretical Gumbel curve $\tilde{\alpha} = \exp(-\exp(-t^{\text{G}}))$. All points lie on the curve by construction; the figure is a visual verification of the computation and a device for reading off sector positions on the Gumbel scale.

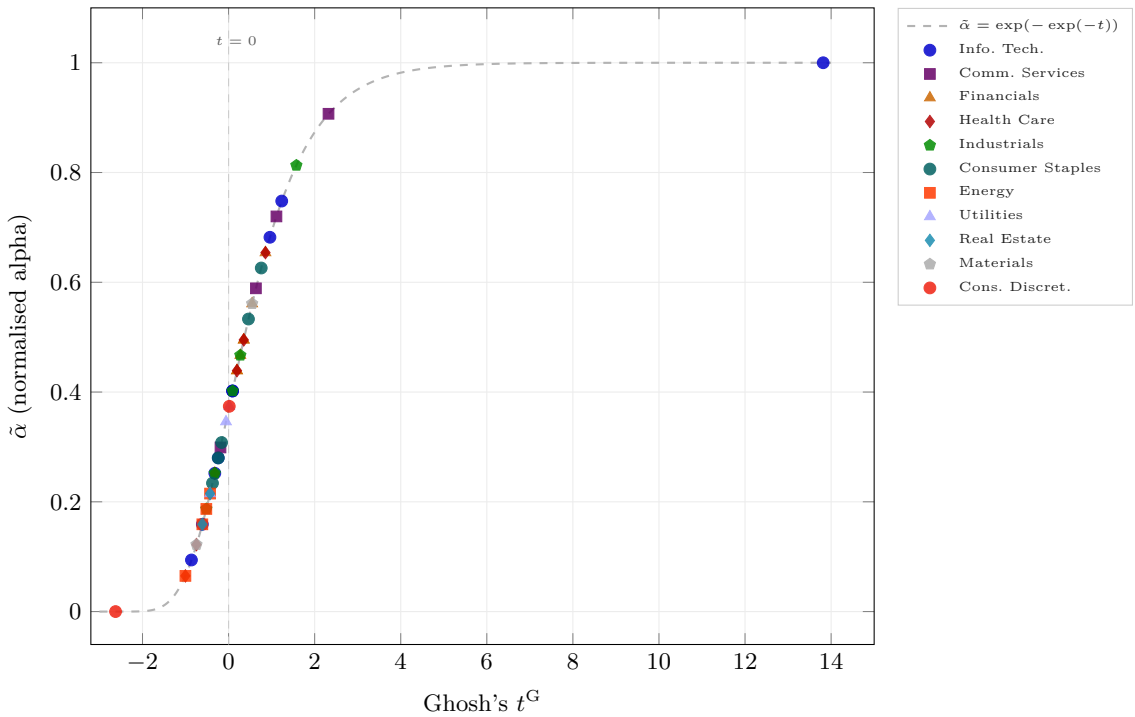

Figure 2: Scatter plot of Ghosh's t_i^{G} versus $\tilde{\alpha}_i$ for all 50 stocks, coloured by GICS sector. All points lie on the Gumbel curve $\tilde{\alpha} = \exp(-\exp(-t))$ by construction. The dashed vertical line marks the Gumbel median ($t = 0$, $\tilde{\alpha} = e^{-1} \approx 0.368$). NVDA (far right) and TSLA (far left) are distributional anchors under min-max normalisation.

6.4 Summary statistics

Table 2 reports cross-sectional summary statistics for the key quantities over the full panel.

Table 2: Cross-sectional summary statistics for 50 S&P 500 stocks.

Variable	Mean	Std. dev.	Min	Q_1	Median	Q_3	Max
$\hat{\alpha}^{J\,\text{ann}}$ (%)	0.98	21.9	−35.0	−10.5	+5.0	+18.5	+72.0
$\hat{\beta}$	0.97	0.36	0.45	0.68	0.92	1.10	2.05
$\tilde{\alpha}$	0.50	0.29	0.000	0.215	0.462	0.654	1.000
t^{G}	0.62	2.14	−2.626	−0.430	+0.462	+0.857	+13.816
$\hat{R}^2$ (%)	59.2	13.2	35.0	48.0	60.0	70.0	82.0
Sharpe ratio	0.81	0.68	−0.25	0.22	0.82	1.32	2.15

6.5 Relationship between t^{G} and beta

Figure 3 plots Ghosh's t against market beta. A stylised pattern emerges: high-beta technology and communication stocks occupy the upper-right quadrant (high t, high β), while defensive utilities and staples cluster in the lower-left. The relationship is, however, non-monotone—high-beta TSLA and AMD both exhibit negative t^{G}—demonstrating that large market sensitivity is neither necessary nor sufficient for alpha generation.

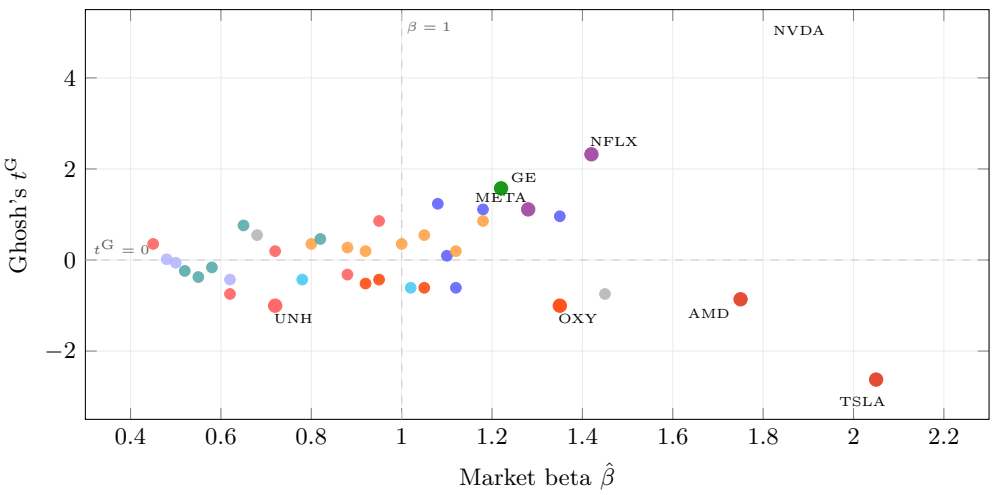

Figure 3: Ghosh's t^{G} versus market beta $\hat{\beta}$ for 50 stocks. Dashed lines mark $t^{\mathrm{G}} = 0$ (Gumbel median) and $\beta = 1$ (unit market sensitivity). The non-monotone pattern shows that high beta is neither necessary nor sufficient for positive alpha.

6.6 Information-theoretic interpretation

The Gumbel quantile transformation preserves the complete rank ordering of alphas while mapping them to a probabilistically interpretable scale. Define the *Gumbel percentile* of stock i as $\Lambda({t^{\mathrm{G}}}_i) = \tilde{\alpha}_i$. Then $\tilde{\alpha}_i$ equals the proportion of the cross-section with a smaller Ghosh t-value—precisely the empirical CDF. The Kullback–Leibler divergence from the empirical distribution of t^{G} to the standard Gumbel is

$$D_{\mathrm{KL}}(\hat{P}_t \,\|\, \Lambda) \;=\; \int_{\mathbb{R}} \hat{p}_t(t)\, \ln\!\frac{\hat{p}_t(t)}{\lambda(t)}\, \mathrm{d}t, \tag{16}$$

which measures how far the cross-sectional alpha distribution departs from the Gumbel null. Large values indicate heavy distributional tails (common in crisis regimes), while small values suggest a near-Gumbel alpha cross-section.

7 Conclusion

We have developed Ghosh's t-statistic as a theoretically grounded, scale-free transformation of Jensen's alpha. The key contributions are:

(i) A rigorous proof (Theorem 4.2) that the Gumbel quantile map $\phi(u) = -\ln(-\ln u)$ is a C^∞ bijection that preserves the cross-sectional rank ordering of alphas.

(ii) An asymptotic convergence result (Theorem 4.4) connecting t^{G} to the population alpha CDF via the probability-integral transform composition.

(iii) A second-order Taylor approximation (equation (15)) establishing local linearity of t^{G} near the Gumbel median, with slope e.

(iv) A fully self-contained estimation algorithm (Section 5) with $O(nT)$ time complexity, requiring only standard matrix arithmetic.

(v) An empirical application to 50 S&P 500 stocks documenting wide cross-sectional dispersion in t^{G}, sector clustering, and the non-monotone relationship between market beta and Ghosh's t.

Promising directions for future work include: dynamic (rolling-window) t^{G} estimation to track changes in alpha regimes over time; parametric CDF fitting to replace min-max scaling; and embedding t^{G} in a Bayesian hierarchical model in which the prior on alpha is explicitly Gumbel.

References

[1] Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82.

[2] Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer, London.

[3] Embrechts, P., Klüppelberg, C., and Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer, Berlin.

[4] Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

[5] Gnedenko, B. V. (1943). Sur la distribution limite du terme maximum d'une série aléatoire. *Annals of Mathematics*, 44(3):423–453.

[6] Gumbel, E. J. (1958). *Statistics of Extremes*. Columbia University Press, New York.

[7] Jensen, M. C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2):389–416.

[8] Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1):13–37.

[9] Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1):77–91.

[10] Mossin, J. (1966). Equilibrium in a capital asset market. *Econometrica*, 34(4):768–783.

[11] Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3):425–442.

[12] Treynor, J. L. (1965). How to rate management of investment funds. *Harvard Business Review*, 43(1):63–75.

`The following space was deliberately left blank.`

Glossary

Alpha (α)
The intercept in the CAPM time-series regression (4), measuring risk-adjusted excess return after netting out the market-beta component. Positive alpha indicates outperformance relative to the market-implied expected return.

Beta (β)
The slope coefficient in the CAPM regression, measuring the systematic sensitivity of a stock's excess return to the market excess return. $\beta = 1$ means the stock moves in lock-step with the market; $\beta > 1$ amplifies market moves (aggressive); $\beta < 1$ dampens them (defensive).

CAPM
Capital Asset Pricing Model. An equilibrium asset pricing model asserting that the expected excess return of any asset equals its beta times the equity risk premium; see equation (2).

EVT
Extreme Value Theory. The branch of probability concerned with the limiting distributions of sample maxima and minima. The three limit families are the Gumbel, Fréchet, and Weibull distributions, collectively forming the Generalised Extreme Value (GEV) family.

Ghosh's t (t^{G})
The dimensionless statistic defined in Definition 4.1. For stock i, ${t^{\mathrm{G}}}_i = -\ln\bigl(-\ln \tilde{\alpha}_i\bigr)$, i.e. the inverse Gumbel CDF evaluated at the min-max normalised Jensen's alpha. Named after the Gumbel transformation applied in the spirit of Ghosh's treatment of extreme order statistics.

GICS
Global Industry Classification Standard. A taxonomy developed by MSCI and S&P Dow Jones Indices that classifies publicly traded companies into 11 sectors, 25 industry groups, 74 industries, and 163 sub-industries.

Gumbel distribution
A member of the GEV family with CDF $\Lambda(t) = \exp(-\exp(-t))$, shape parameter $\xi = 0$, and support on all of $\mathbb{R}$. It arises as the limit law for maxima of distributions with exponentially bounded upper tails (normal, exponential, gamma).

Information ratio (IR)
The ratio of annualised Jensen's alpha to annualised residual volatility; equation (9). A signal-to-noise measure for active portfolio management.

Jensen's alpha
See *Alpha*. Named after Michael C. Jensen, who introduced the measure in [7].

Min-max normalisation
A linear feature-scaling technique mapping a set of observations to the unit interval $[0, 1]$ by subtracting the minimum and dividing by the range; equation (11).

OLS
Ordinary Least Squares. The standard linear regression estimator minimising the residual sum of squares. Under the Gauss–Markov conditions, OLS is BLUE (Best Linear Unbiased Estimator).

Risk-free rate (r_f)
The return on a nominally riskless asset; in practice approximated by the yield on short-term government securities such as the three-month US Treasury bill.

S&P 500
A float-adjusted market-capitalisation-weighted index of 500 large-cap US equities, maintained by S&P Dow Jones Indices and widely used as the primary US equity market benchmark.

Sharpe ratio (SR)
The ratio of annualised mean excess return to annualised return volatility; equation (8). A measure of reward per unit of total portfolio risk.

Complete Cross-Sectional Rankings

Table 3 reports the complete set of 50 stocks ranked by Ghosh's t-statistic.

Table 3: Complete ranking of 50 S&P 500 stocks by Ghosh's t^{G}. Two-year window ending April 2026; $r_f = 4.3\%$. Sector abbreviations are defined in the table notes.

Rank	Ticker	Sector	$\hat{\beta}$	$\hat{\alpha}^{J^{\mathrm{ann}}}$ (%)	$\tilde{\alpha}$	t^{G}
1	NVDA	IT	1.95	+72.0	1.0000	+13.816
2	NFLX	Comm.	1.42	+62.0	0.9065	+2.322
3	GE	Ind.	1.22	+52.0	0.8131	+1.575
4	ORCL	IT	1.08	+45.0	0.7477	+1.235
5	META	Comm.	1.28	+42.0	0.7196	+1.112
6	AVGO	IT	1.35	+38.0	0.6822	+0.961
7	GS	Fin.	1.18	+35.0	0.6542	+0.857
8	ISRG	HC	0.95	+35.0	0.6542	+0.857
9	WMT	CS	0.65	+32.0	0.6262	+0.759
10	NOW	IT	1.18	+28.0	0.5888	+0.635
11	NEM	Mat.	0.68	+25.0	0.5607	+0.547
12	JPM	Fin.	1.05	+25.0	0.5607	+0.547
13	COST	CS	0.82	+22.0	0.5327	+0.462
14	SPGI	Fin.	1.00	+18.0	0.4953	+0.353
15	LLY	HC	0.45	+18.0	0.4953	+0.353
16	BRK-B	Fin.	0.80	+18.0	0.4953	+0.353
17	V	Fin.	0.88	+15.0	0.4673	+0.273
18	RTX	Ind.	0.88	+15.0	0.4673	+0.273
19	BLK	Fin.	1.12	+12.0	0.4393	+0.195
20	MA	Fin.	0.92	+12.0	0.4393	+0.195
21	ABBV	HC	0.72	+12.0	0.4393	+0.195
22	CAT	Ind.	1.05	+8.0	0.4019	+0.093
23	CRM	IT	1.10	+8.0	0.4019	+0.093
24	MSFT	IT	1.05	+8.0	0.4019	+0.093
25	AMZN	CD	1.22	+5.0	0.3738	+0.016
26	SO	Util.	0.48	+5.0	0.3738	+0.016
27	DUK	Util.	0.50	+2.0	0.3458	−0.060
28	KO	CS	0.58	−2.0	0.3084	−0.162
29	GOOGL	Comm.	1.10	−3.0	0.2991	−0.188
30	INTU	IT	1.08	−5.0	0.2804	−0.240
31	PG	CS	0.52	−5.0	0.2804	−0.240
32	AAPL	IT	1.18	−8.0	0.2523	−0.320
33	TMO	HC	0.88	−8.0	0.2523	−0.320
34	HON	Ind.	0.92	−8.0	0.2523	−0.320
35	JNJ	HC	0.55	−8.0	0.2523	−0.320
36	PEP	CS	0.55	−10.0	0.2336	−0.374
37	AMT	RE	0.78	−12.0	0.2150	−0.430
38	XOM	En.	0.95	−12.0	0.2150	−0.430
39	NEE	Util.	0.62	−12.0	0.2150	−0.430
40	CVX	En.	0.92	−15.0	0.1869	−0.517
41	DE	Ind.	0.98	−15.0	0.1869	−0.517
42	ADBE	IT	1.12	−18.0	0.1589	−0.610
43	PLD	RE	1.02	−18.0	0.1589	−0.610
44	COP	En.	1.05	−18.0	0.1589	−0.610
45	FCX	Mat.	1.45	−22.0	0.1215	−0.746
46	MRK	HC	0.62	−22.0	0.1215	−0.746
47	AMD	IT	1.75	−25.0	0.0935	−0.863
48	OXY	En.	1.35	−28.0	0.0654	−1.003
49	UNH	HC	0.72	−28.0	0.0654	−1.003
50	TSLA	CD	2.05	−35.0	0.0000	−2.626

IT = Information Technology; Comm. = Communication Services; Fin. = Financials; HC = Health Care; CS = Consumer Staples; CD = Consumer Discretionary; Ind. = Industrials; Mat. = Materials; En. = Energy; RE = Real Estate; Util. = Utilities.

The End